

Abdul Khurram

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EDUCATION & AWARDS

University of British Columbia

Vancouver, BC, Canada

Bachelor of Science in Computer Science (*Expected Graduation: May 2027*)

Euclid Mathematics Contest, University of Waterloo

2021

Top 5% Internationally, Certificate of Distinction

WORK EXPERIENCE

Software Developer Co-op | *Python, Apache Spark, Airflow, SQL, Linux*

Jan. 2026 – Present

RBC Royal Bank

Vancouver, BC, Canada

- Engineered **Python**/Airflow **ETL pipelines** powering **ML** workloads across 50M+ daily transactions on petabyte-scale Delta Lake, delivering feature-ready datasets for **model training** and real-time inference.
- Optimized **Spark algorithms** via Spark-native API migration and AQE/cluster-memory tuning, cutting latency 40% and compute costs through **distributed computing** resource profiling.
- Hardened batch processing framework with observability, fault tolerance, and automated failover; shipping cost/performance dashboards driving real-time infrastructure decisions across the platform.

Software Engineer Intern | *Python, TypeScript/Node.js, AWS, OpenAI*

Jan. – Dec. 2025

Contello.ai

Toronto, ON, Canada

- Integrated OpenAI Responses API into production TypeScript/Node.js on AWS; built content-generation API services serving thousands/day with **model inference** observability via CloudWatch.
- Enabled **ML**-assisted content creation by ingesting brand docs via an **NLP** pipeline, caching **embeddings**, and auto-injecting context while enforcing responsible AI guardrails.
- Raised service reliability to 99.9% via **model** API migration, cost/latency tracking, retry/backoff, and automated tests in CI/CD; deployed via Docker containers on AWS.

RESEARCH EXPERIENCE

Software Developer & Team Lead | *Unreal Engine, C++, Python*

Sept. – Dec. 2024

UBC Emerging Media Lab

Vancouver, BC, Canada

- Led a **human-AI collaboration research** project: built an AI-driven VR simulation in Unreal (**C++**) pairing nurse practitioners with AI patients for clinical training.
- Designed a multithreaded supervisory architecture (**C++/Python**) enabling real-time **ML model** monitoring and protocol enforcement across autonomous patient actors.
- Optimized** inference pipeline, achieving 40% latency reduction and 90% cost savings through profiling and **algorithm** tuning; documented design decisions and system capabilities.

PROJECTS

CircleFund | *Python, PyTorch, TypeScript/Next.js, Prisma/PostgreSQL*

Personal Project

- Building a full-stack lending platform (Next.js, Prisma/PostgreSQL) with account onboarding, invite-code circle management, and contribution dashboards for community-group financial workflows.
- Designing **PyTorch**-based borrower risk scoring and contribution reliability **models** with planned liquidity stress forecasting, integrating **ML** predictions into the platform's decision layer.

Backer | *TypeScript/Next.js, AWS DynamoDB, NextAuth*

Hack The Coast 2026

- Architected a startup discovery platform (TypeScript/Next.js, DynamoDB) serving personalized investor feeds at scale via API services.
- Built **ML** training pipeline: supervised **model** with sequence and clustering features, ingesting signals from DynamoDB event streams to train and deploy an **XGBoost** ranker to production.

TECHNICAL SKILLS

Languages: Python, Java, SQL, TypeScript/JavaScript, C/C++, Bash, HTML/CSS

ML & Platforms: PyTorch, Scikit-learn, OpenAI API, LangChain, NLP/embeddings; React/Next.js, Airflow, Hadoop/Spark; AWS (Lambda, DynamoDB, S3, SQS, CloudWatch); PostgreSQL, Docker, Git, CI/CD